Regression 2

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Topics for this lecture

- Predictions (with uncertainty)
- Significance of regression, the F-test
- Interaction effects
- Model checking
- Polynomial regression
- Transformations of the response
- Variable selection

The body data

Let's again look at the body data and the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \epsilon_i$$

with the usual assumption of independent errors distributed as $\epsilon_i \sim N(0, \sigma^2)$. Here y_i is the body weight of person i, x_{1i} the height, x_{2i} is the circumference and x_{3i} is the age of the same person.

Our model with 3 predictors

```
##
## Call:
## lm(formula = Weight ~ Height + Circumference + Age, data = bodydata,
##     subset = 1:20)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.4088 -3.3900 -0.9123  3.0403  6.9827
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -66.5785    30.0098  -2.219 0.041330 *
## Height          0.4768     0.1600   2.981 0.008831 **
## Circumference   0.7769     0.1858   4.181 0.000706 ***
## Age            -0.2094     0.3731  -0.561 0.582352
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.259 on 16 degrees of freedom
## Multiple R-squared:  0.6852, Adjusted R-squared:  0.6262
## F-statistic: 11.61 on 3 and 16 DF,  p-value: 0.0002729
```

Prediction

A fitted model can be used for prediction of new observations

According to model 4, what is the predicted Weight of some person with these values?

```
##   Height Circumference Age
## 1    171            76  30
```

$$\hat{y} = -66.579 + 0.4768 \cdot 171 + 0.7769 \cdot 76 - 0.2094 \cdot 30$$

$$= 67.72$$

```
newobs <- data.frame('Height'=171, 'Circumference'=76, 'Age'=30)
predict.lm(LinearModel.4, newdata=newobs)
```

```
##       1
## 67.7196
```

Confidence in predictions

Confidence interval for
$E(Y|x_1, \ldots, x_p) \in \widehat{\mu}(\mathbf{x}_{\text{new}}) \pm t_{\alpha/2,\, n-p-1} \cdot \text{SE}(\widehat{\mu}(\mathbf{x}_{\text{new}}))$ where $\text{SE}(\widehat{\mu}(\mathbf{x}_{\text{new}}))$ accounts for uncertainty in estimation.

```
predict.lm(LinearModel.4, newobs, level=0.95, interval=c("confidence"))
```

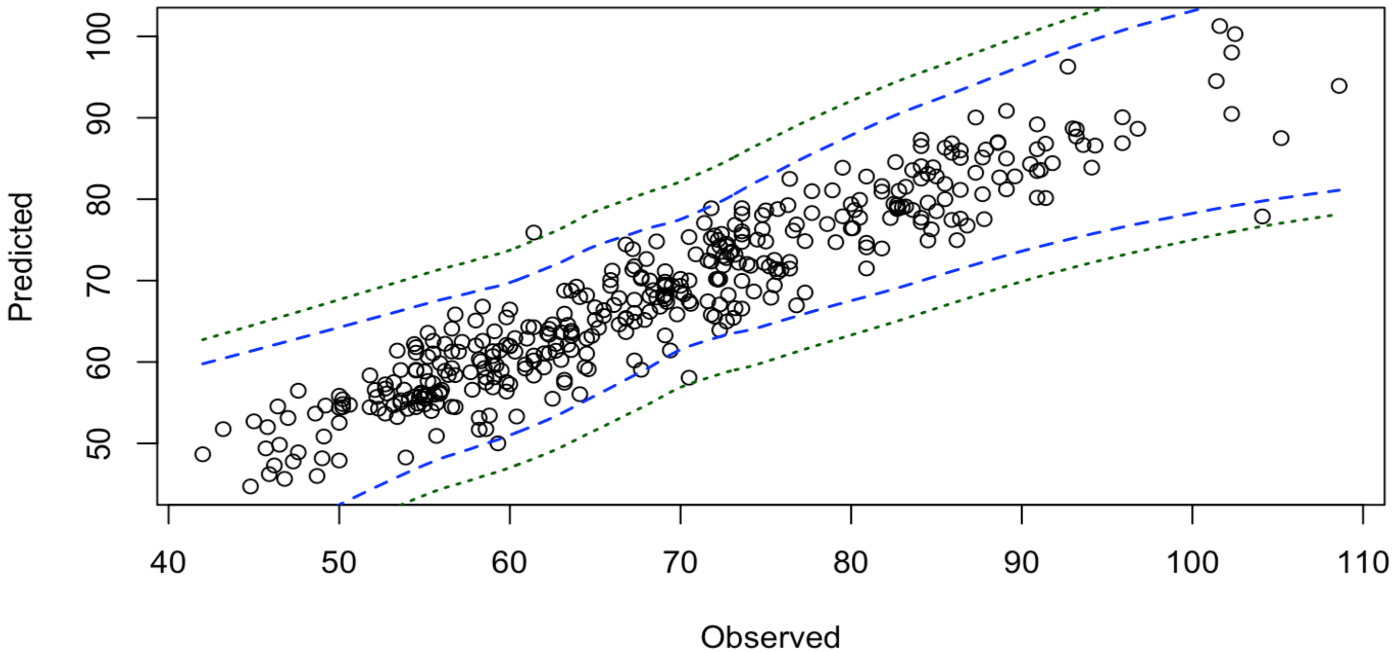
```
##        fit      lwr      upr
## 1 67.7196 60.98252 74.45668
```

Prediction interval for $Y|x_1, \ldots, x_p \in \widehat{\mu}(\mathbf{x}_{\text{new}}) \pm t_{\alpha/2,\, n-p-1}\, \text{SE}_{\text{pred}}$

```
predict.lm(LinearModel.4, newobs, level=0.95, interval=c("prediction"))
```

```
##        fit      lwr      upr
## 1 67.7196 56.45463 78.98456
```

Predicted vs observed - plot



Check predictions' errors

```
##         fit      lwr      upr
## 25 73.94436 64.06824 83.82047
## 26 82.80148 72.72603 92.87693
## 27 81.03059 71.44393 90.61725
## 28 71.46852 61.81893 81.11811
## 29 65.93065 55.27452 76.58677
## 31 76.31124 66.90951 85.71296
## 32 69.35871 59.61763 79.09979
## 33 75.73802 65.67258 85.80346
## 34 82.52507 72.84459 92.20555
## 35 67.71321 57.86212 77.56431
## 37 73.86325 64.29154 83.43496
## 39 74.81258 65.46899 84.15617
```

```
## [1] " RMSE  observed: 3.8092"
```

```
## [1] " RMSE  predictions: 4.8026"
```

Checking the prediction intervals

How many observations fell within the prediction intervals?

```
largerthanlower <- (newWeight > pred[,2])
smallerthanupper <- (newWeight < pred[,3])
inInterval <- which(largerthanlower & smallerthanupper)
print(length(inInterval))
```

```
## [1] 381
```

Proportion of observations within prediction intervals:

```
length(inInterval)/length(newWeight)
```

```
## [1] 0.9844961
```

What should we expect?

SSE, SST, SSR again

- Total Sum of Squares: $SST = \sum_{i=1}^{n} (y_i - \bar{y})^2$
- Error Sum of Squares: $SSE = \sum_{i=1}^{n} (y_i - \widehat{\mu}(x_i))^2$
- Regression Sum of Squares: $SSR = \sum_{i=1}^{n} (\widehat{\mu}(x_i) - \bar{y})^2$

  It can be shown:

$$SST = SSE + SSR \quad \Rightarrow \quad R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

Significance of regression?

We have seen how we can test the significance of single predictor terms with T-tests in the previous lesson.

We may also test the overall significance of the regression model.

Recall from earlier that $R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$ is a measure of model fit. If the model can explain much of the observed variability in y, then SSR is close to SST and big compared to SSE.

Can be shown that a convenient test observator, typically named F (for Fisher) is:

$$F = \frac{SSR/p}{SSE/(n - p - 1)}$$

The F-test for overall significance of regression

Hypothesis:

$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0$ vs H_1 : at least one $\beta_j \neq 0$.

Reject the null hypothesis at test level α if F is large, and more specifically if

$$F > F_{\alpha,p,n-p-1}$$

Let's see how this turns out for the body data model.

The F-test illustration

The body data

```
body1 <- lm(Weight ~  Height + Circumference + Age, data=bodydata, subset = 1:20)
Atab <- anova(body1)
```

```
## Analysis of Variance Table
##
## Response: Weight
##               Df Sum Sq Mean Sq F value    Pr(>F)
## Height         1 308.31  308.31 16.9984 0.0007973 ***
## Circumference  1 317.76  317.76 17.5190 0.0006993 ***
## Age            1   5.72    5.72  0.3151 0.5823519
## Residuals     16 290.20   18.14
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##  SST =  921.988 , SSR =  631.7836 ;
##  Cummulative SST 308.3131 626.0686 631.7836 921.988 ;
##  F =  11.61083
```

The body data - Compare with previous

```
body1 <- lm(Weight ~  Height + Circumference + Age, data=bodydata, subset = 1:20)
Atab <- anova(body1)
```

```
## Analysis of Variance Table
##
## Response: Weight
##               Df Sum Sq Mean Sq F value    Pr(>F)
## Circumference  1 467.18  467.18 25.7575 0.0001125 ***
## Height         1 158.89  158.89  8.7599 0.0092221 **
## Age            1   5.72    5.72  0.3151 0.5823519
## Residuals     16 290.20   18.14
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## SST =  921.988 , SSR =  631.7836 , F =  11.61083
```

Interaction effects

Interaction effects occur when the effect of one variable on the response variable depends on the level/value of another variable.

Consider the model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{2i} x_{3i} + \epsilon_i$$

here we have expanded the model with an interaction term between x_2 and x_3 (between circumference and age). How should we interpret this model?

How to fit the model in RStudio

Now fit on all the data we have

```
data(bodydata)
body2 <- lm(Weight ~ Height + Circumference + Age +
              Circumference:Age, data=bodydata)
summary(body2)
```

The output with all data

```
##
## Call:
## lm(formula = Weight ~ Height + Circumference + Age + Circumference:Age,
##     data = bodydata)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1735  -2.5820  -0.1732   2.6033  22.5148
##
## Coefficients:
##                     Estimate Std. Error t value Pr(>|t|)
## (Intercept)       -93.579875   6.030390 -15.518  < 2e-16 ***
## Height              0.414057   0.027243  15.199  < 2e-16 ***
## Circumference       1.244437   0.070990  17.530  < 2e-16 ***
## Age                 0.611383   0.159085   3.843 0.000141 ***
## Circumference:Age  -0.009476   0.001990  -4.763 2.67e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.252 on 402 degrees of freedom
## Multiple R-squared:  0.8989, Adjusted R-squared:  0.8979
## F-statistic: 893.8 on 4 and 402 DF,  p-value: < 2.2e-16
```

Model checking

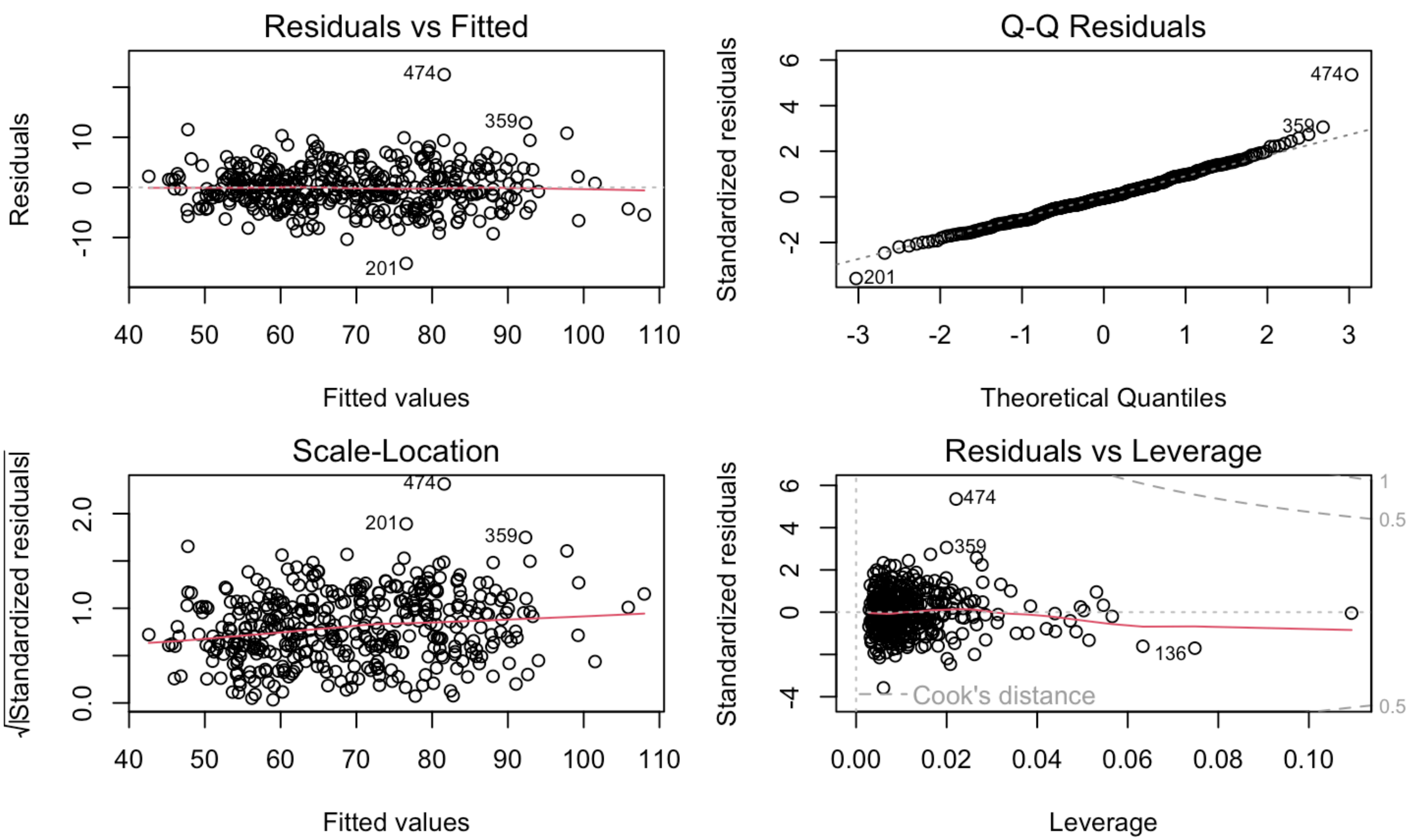
In standard linear regression we typically assume:

- Linearity
- Constant variance
- Independent observations
- Normality

We should check that these assumptions are OK. Typically we do this by examining the residuals

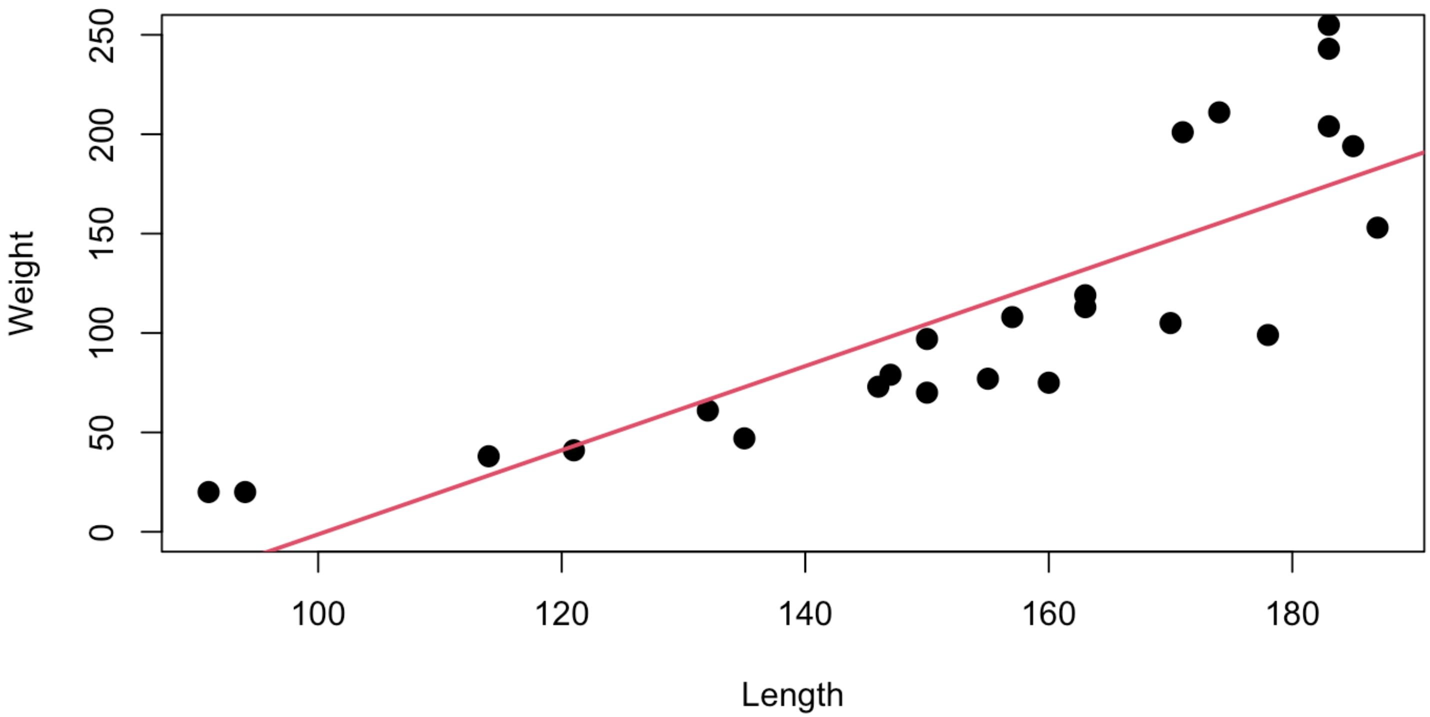
Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))
plot(body2, ask=FALSE)
```



Draw Q-Q plot and std residuals

Another example - Bears

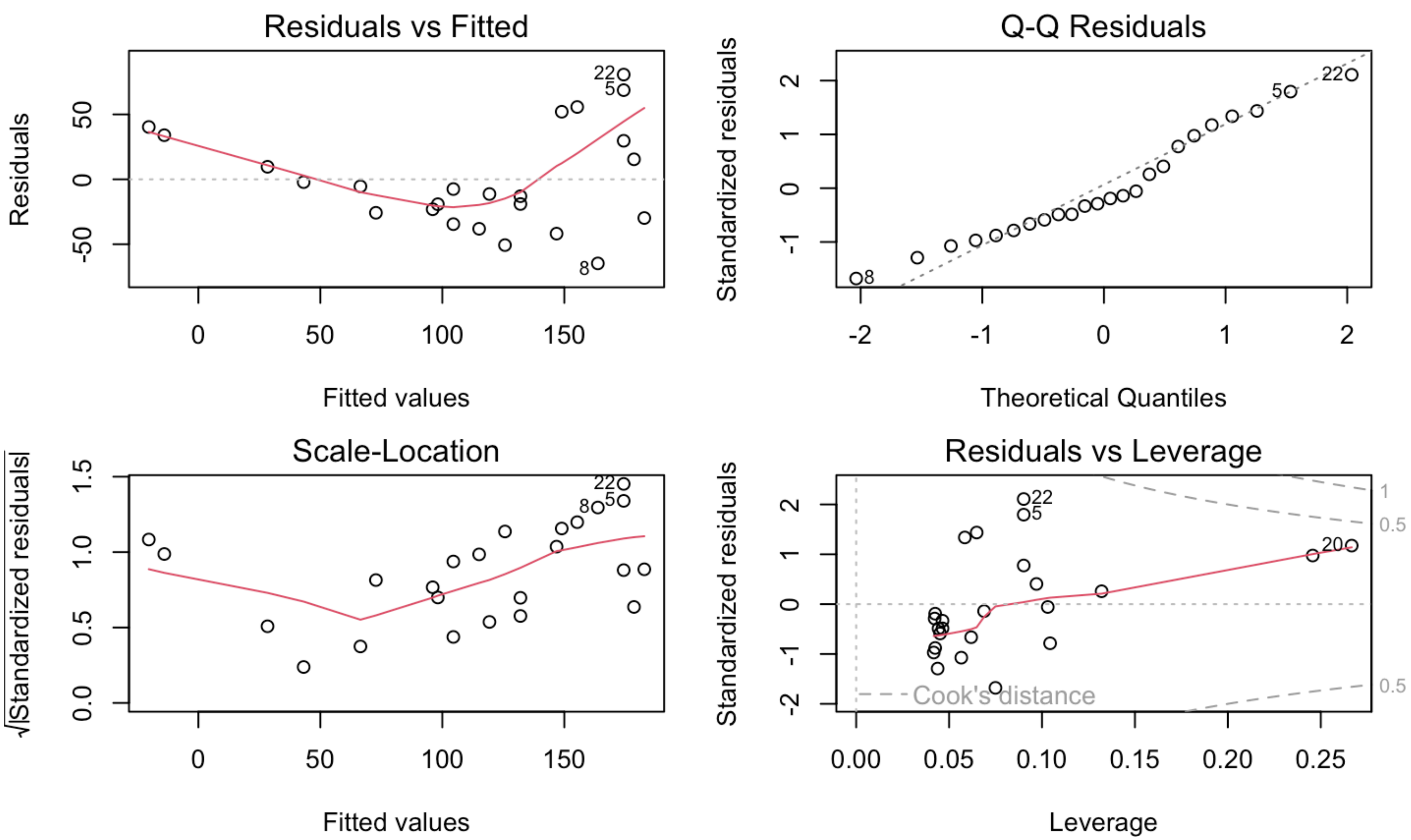


Summary

```
##
## Call:
## lm(formula = Weight ~ Length, data = bears)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -64.76 -26.78  -9.42  30.74  80.67
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -212.8522    47.2818  -4.502 0.000177 ***
## Length         2.1158     0.3027   6.989 5.15e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 40.1 on 22 degrees of freedom
## Multiple R-squared:  0.6895, Adjusted R-squared:  0.6754
## F-statistic: 48.85 on 1 and 22 DF,  p-value: 5.147e-07
```

Residual plots

```r
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))
plot(bears1, ask=FALSE)
```



Outliers and influence statistics

- Leverage
- Standardized residuals
- Deleted residuals (PRESS residuals) (predict the deleted leave out one point)
- Cook’s distance (combines leverage and standardized residuals) > 1, can be dangerous

Explain leverage

Polynomial regression

Non-linear relationships may be modelled by polynomial regression.

Bear example

- $y =$ Weight
- x_1 = Length
- x_1^2 = Length^2

A multiple LINEAR model (linear in the β's)

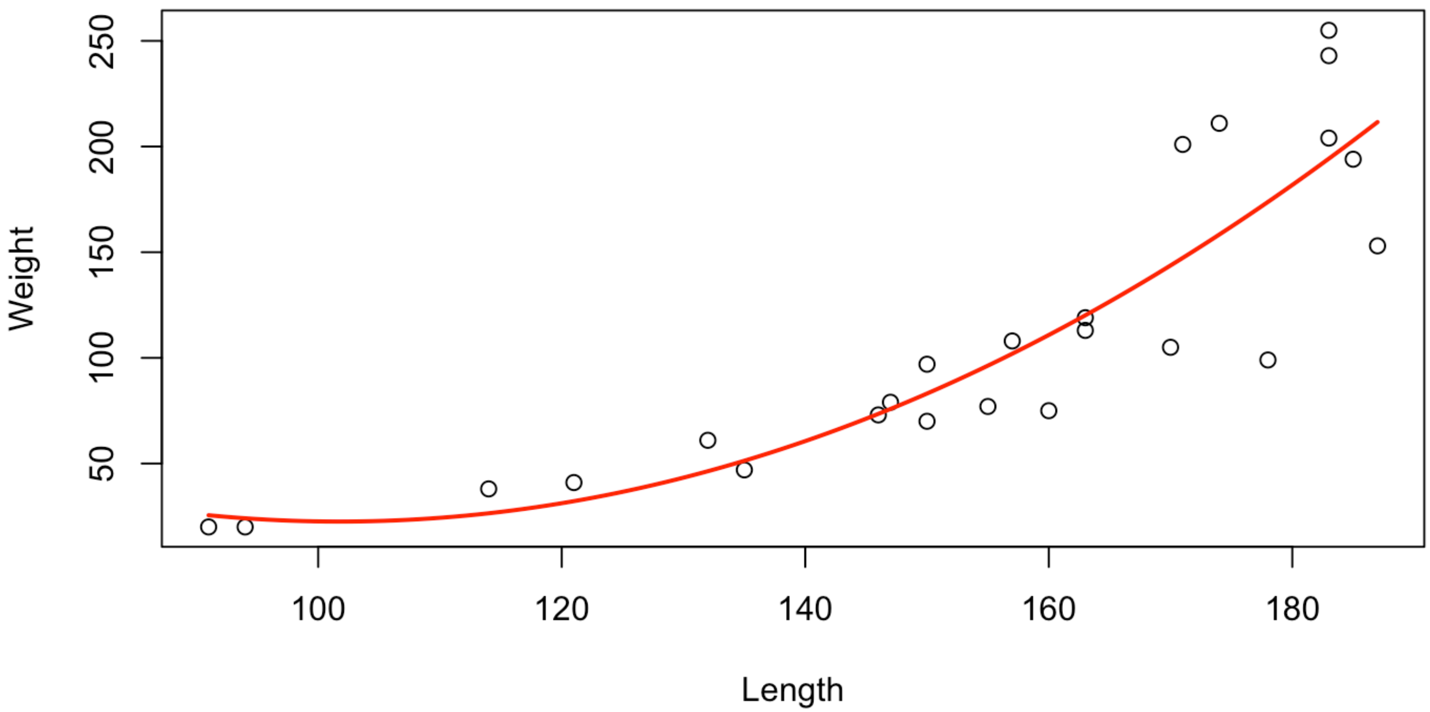
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon$$

Fitting the model in R

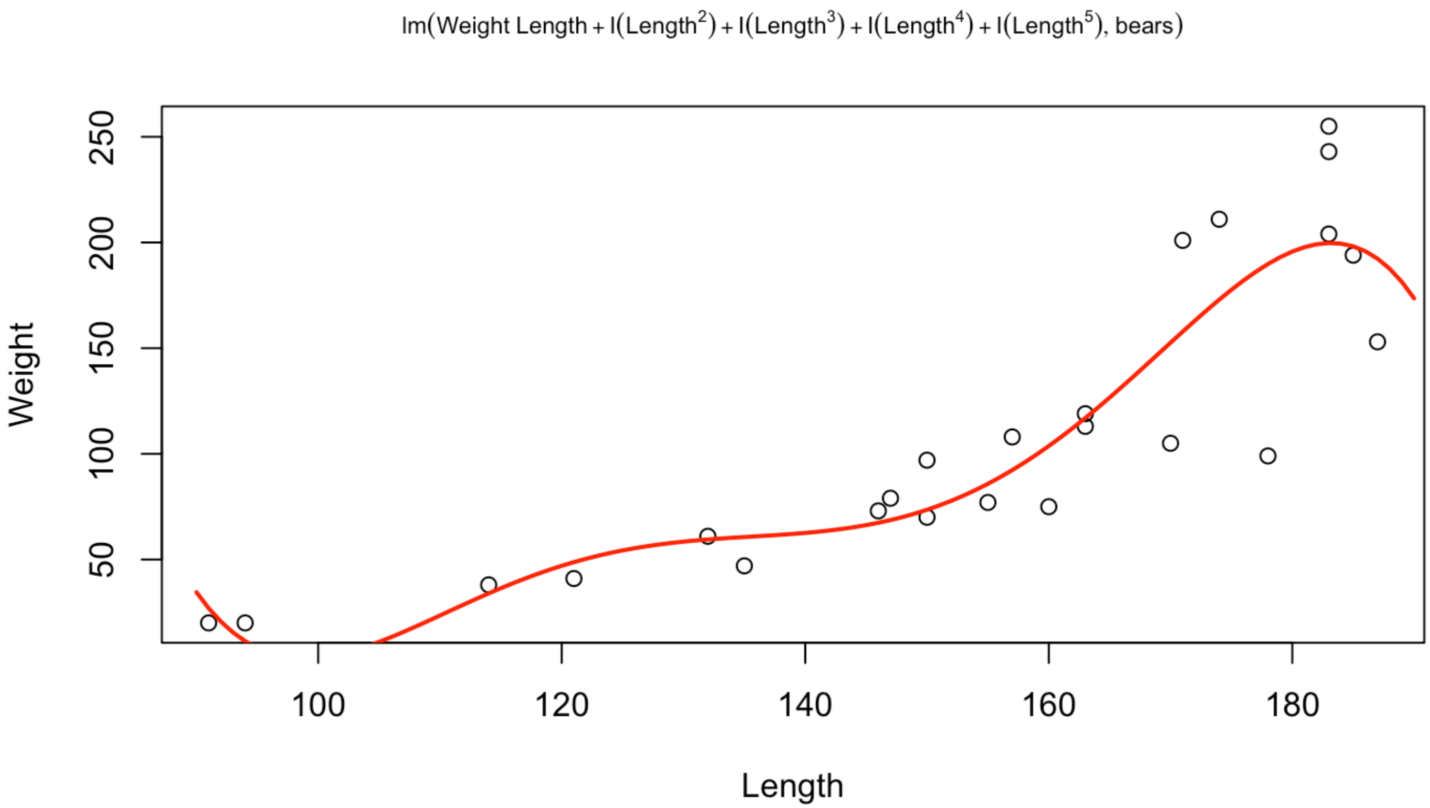
```
data(bears)
bears2 <- lm(Weight ~ Length + I(Length^2), data=bears)
summary(bears2)
```

```
##
## Call:
## lm(formula = Weight ~ Length + I(Length^2), data = bears)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -75.411 -10.514  -1.302  11.809  60.075
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept) 291.623760 175.892266   1.658  0.11219
## Length       -5.289645   2.525406  -2.095  0.04852 *
## I(Length^2)   0.026018   0.008825   2.948  0.00768 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 34.52 on 21 degrees of freedom
## Multiple R-squared:  0.7804, Adjusted R-squared:  0.7595
```

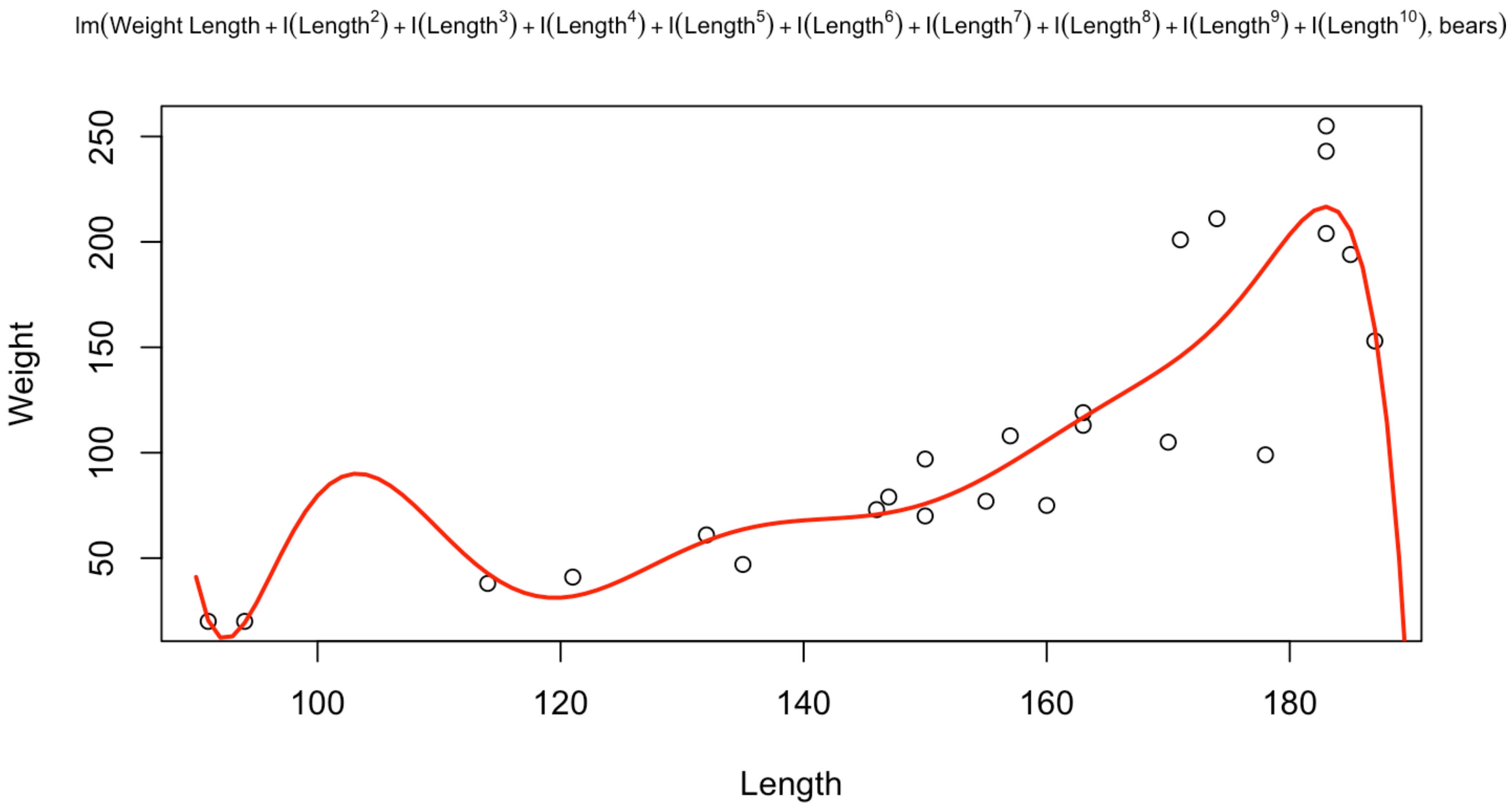
A fitted line plot



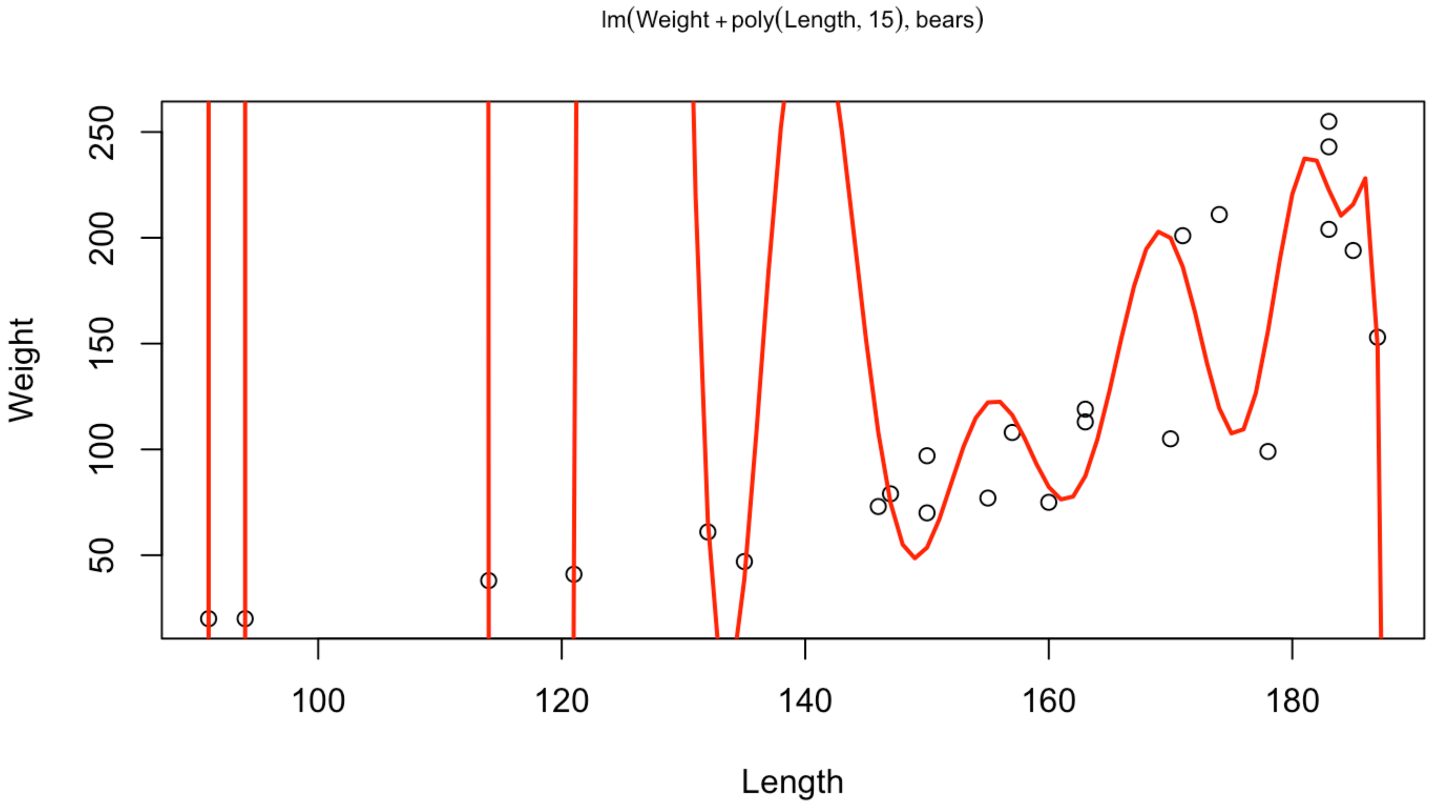
Careful! Over-fitting!



Careful! Over-fitting!



Careful! Over-fitting!



Transformations

Deviations from model assumptions may be fixed by transforming y and/or x.

Some transformations which may improve linearity and stabilize variance are

- $\tilde{y} = log(y)$
- $\tilde{x} = log(x)$
- $\tilde{y} = \sqrt{y}$

Then use y' and/or x' instead of y and x to fit the model

The log(Weight) response model

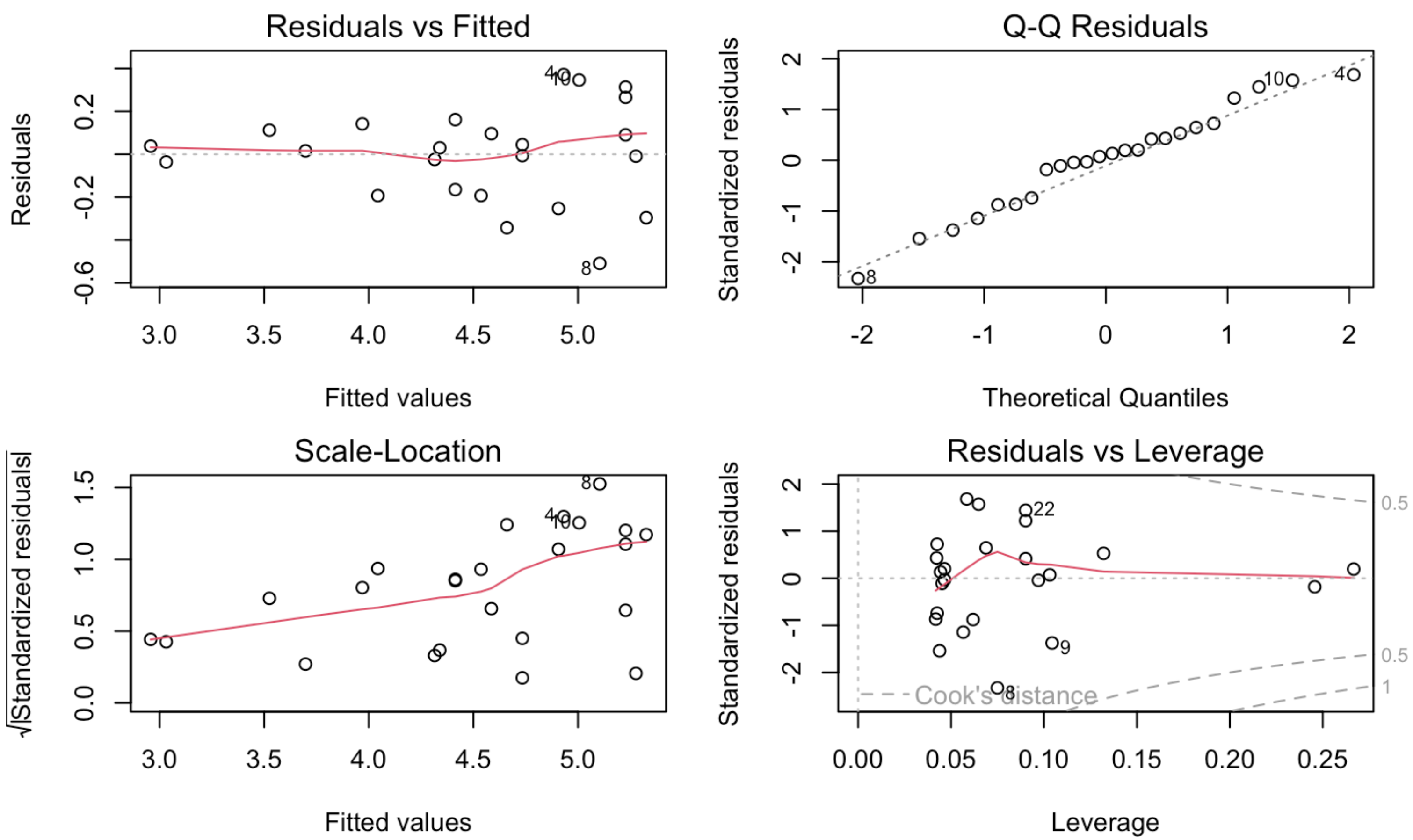
```
bears4 <- lm(log(Weight) ~ Length, data=bears)
summary(bears4)
```

```
##
## Call:
## lm(formula = log(Weight) ~ Length, data = bears)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.50918 -0.17192  0.02291  0.11978  0.37173
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 0.712120   0.268342   2.654   0.0145 *
## Length      0.024675   0.001718  14.363 1.17e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2276 on 22 degrees of freedom
## Multiple R-squared:  0.9036, Adjusted R-squared:  0.8992
## F-statistic: 206.3 on 1 and 22 DF,  p-value: 1.173e-12
```

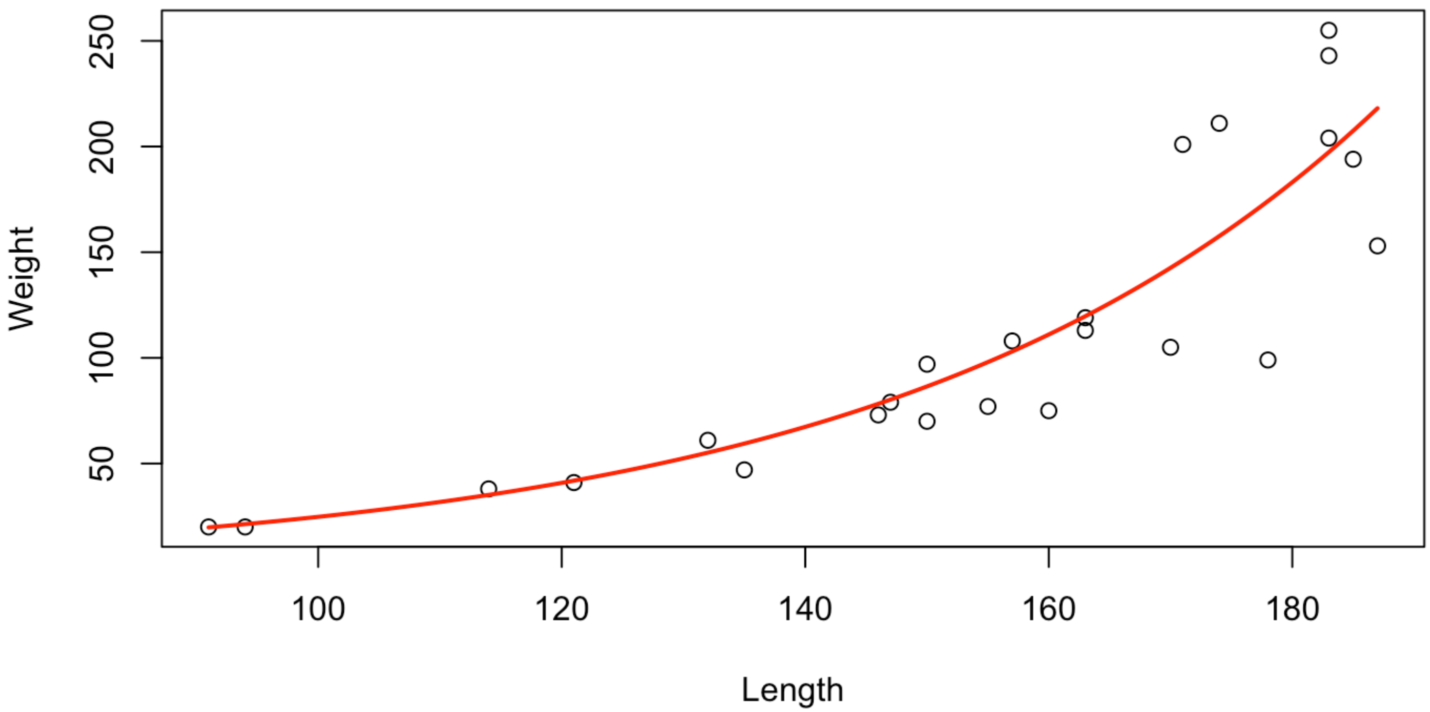
Explain the transformation and back-transformation

Residual plots

```
par(mfrow=c(2,2), mar=c(4.5,4,1.5,1.5))
plot(bears4, ask=FALSE)
```



A fitted line plot



Variable selection

Variable selection is the process of selecting a subset of variables that are most relevant for the response in some way. Various approaches. Some are based on hypothesis testing, some on criteria (like Akaike's Information Criteria - AIC, Bayesian information criterion - BIC, adjuseted R^2_{adj}, etc)

Sometimes combined with stepwise approaches like:

- Forward selection
- Backward selection
- Forward/Backward
- Backward/Forward
- All subsets selection

```
library(mixlm)
```

```
##
## Attaching package: 'mixlm'
```

Stepwise methods - Backward selection

```r
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata, subset=1:50)
backward(body1, alpha=0.05)
```

```
## Backward elimination, alpha-to-remove: 0.05
##
## Full model: Weight ~ Height + Circumference + Age
##
##     Step    RSS    AIC  R2pred     Cp F value Pr(>F)
## Age    1 964.52 153.98 0.71627 3.3702  1.3702 0.2478
```

```
##
## Call:
## lm(formula = Weight ~ Height + Circumference, data = bodydata,
##     subset = 1:50)
##
## Coefficients:
##   (Intercept)         Height  Circumference
##      -88.7749         0.4363         1.0916
```

Stepwise methods - Forward selection

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata, subset=1:50)
fs <- forward(body1, alpha=0.05)
```

```
## Forward selection, alpha-to-enter: 0.05
##
## Full model: Weight ~ Height + Circumference + Age
##
##               Step     RSS    AIC  R2pred      Cp F value    Pr(>F)    
## Circumference    1 1382.37 169.98 0.60830 21.8918  87.788 2.043e-12 ***
## Height           2  964.52 153.98 0.71627  3.3702  20.361 4.281e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Best subsets method

If the number of candidate variables, p, is limited, e.g. ≤ 20, we may compare all possible models

There are 2^p possible models

Compare all model fits by AIC, BIC, R^2_{adj}

Choose model based on:

- Best-fit - simplicity trade-off

Best subsets method - Example

```
body1 <- lm(Weight ~ Height + Circumference + Age, data=bodydata, subset=1:50)
bs <- best.subsets(body1)
```

```
##          Height Circumference Age       RSS         R2      R2adj         Cp
## 1  ( 1 )                    *       13731.668 0.80903505 0.80856354  317.73818
## 1  ( 2 )      *                     34547.127 0.51955652 0.51837024 1410.28546
## 1  ( 3 )                        *   69391.877 0.03497401 0.03259123 3239.19241
## 2  ( 1 )      *             *        8285.670 0.88477201 0.88420157   33.89246
## 2  ( 2 )                    *   *   12297.068 0.82898589 0.82813929  244.43992
## 2  ( 3 )      *                 *   32876.376 0.54279148 0.54052808 1324.59227
## 3  ( 1 )      *             *   *    7678.048 0.89322215 0.89242728    4.00000
```